

KONRAD ZUSE

Born in Berlin, German engineer Zuse designed and built a programmable computer while working in complete isolation, making the prototype machine in his parents' apartment.



Zuse (1910–1995) conceived the idea of a computer to automate the laborious calculations required in his early career as a civil engineer. In 1936, he left his job at an aviation firm and began constructing his computer, creating the Z1 in 1938. Zuse worked for the German military during World War II, completing the Z3—the world's first fully programmable working computer—in 1941. He went on to develop one of the first high-level programming languages, "Plankalkül."

potential for the quicker electronic technology to realize his 1936 idea of a universal machine. With the war over, Turing worked for the UK's National Physical Laboratory from 1945, where he submitted detailed plans for an Automatic Computing Engine (ACE) that could be programmed via instructions stored in an electronic memory. This meant that unlike the cumbersome Colossus, which had to be rewired to perform each new task, ACE could switch from numerical work to algebra or chess-playing simply by accessing a different program in its

memory, just like a modern computer. Much to Turing's frustration, it was never built, though the 1950 Pilot ACE was a smaller version of his design.

Machine intelligence

After moving to Manchester University in 1948, Turing led the programming team for the Ferranti Mark I—the world's first commercially available digital electronic computer. While there, he explored how machines could be said to think, using a thought experiment in which a computer tries to convince an interviewer that it is human: success meant it was intelligent. The Turing test, as it became known, was influential in Artificial Intelligence, an emerging field pioneered in the US.

Turing also demonstrated a remarkable range of academic interests. He was fascinated by neurology and biology and wrote a 1952 paper on mathematical morphogenesis, the theory of how form and pattern—such as a tiger's stripes—arise in nature.

In the same year, Turing was convicted of being sexually involved with a man, which was illegal at the time. In 1954, at the age of 41, he committed suicide. In 2013, Turing was granted a royal pardon.

DEvised A METHOD
FOR ASSESSING
ARTIFICIAL
INTELLIGENCE

CREDITED WITH TAKING
2 YEARS
OFF THE
DURATION
OF WORLD WAR II



IT IS ESTIMATED THAT
CRACKING
ENIGMA
SAVED MORE THAN
14 MILLION LIVES